

# The evolution of the user experience: computer graphics and interactivity

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## CCS CONCEPTS

• **Human-centered computing**; • **Computing methodologies**  
→ **Computer graphics**;

## KEYWORDS

interactive graphics, human-computer interaction, user experience

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## 1 INTRODUCTION

In this retrospective panel, we dive into the evolution and the place of user experience in computer graphics, a fascinating area that is the foundation of how humans and the digital interact. We bring together distinguished panelists, all SIGGRAPH veterans, to talk about their stories in growing the interactive graphics experience across several decades and across the world. They include pioneers in interactive computer graphics from North America, Europe and Asia who discuss their diverse perspectives and practice in research, academia and industry. They discuss their paths and their insights in the evolution of the user experience in computer graphics through their careers, and how the SIGGRAPH research focus shifted away from the user and needs to come back. Thus, we present this panel as both inspiration and provocation for young SIGGRAPH researchers to bring a human-centered perspective into their work in crafting the future.

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\*Moderator

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## 2 THE PANEL TOPICS

This Retrospective panel is about the evolution of human interaction with computer graphics, and how it grew - or diverged from - from an early SIGGRAPH focus on interactive techniques to the central question of how humans experience, reason with, and operate in a interactive computer graphics world. Our four panelists all began their research in different backgrounds, but they all converge on the essential challenges of better understanding and supporting the human. James Foley literally wrote the book on computer graphics and is considered one of the fathers of human-computer interaction. Gudrun Klinker is a seminal scholar in developing immersive XR approaches for large-scale industrial applications where user experience is key. Kiyoshi Kiyokawa's substantial contributions in early AR/VR techniques led him to a focus on human augmentation and assistive interfaces, envisioning a society in which advanced computer graphics technologies enable a more inclusive society. Maureen Stone's long research career in digital color, interactive techniques and human perception grew to a broader focus on the role of design in how people think, interact and work with visual interfaces.

The panel begins with a brief introduction and position statement from each panelist detailing how they came to to their diverse lenses of understanding and working with the importance of the user experience in computer graphics. However, most of the panel is devoted to a lively round table discussion of their insights and stories around the following topics:

- In the early years of SIGGRAPH, interaction played a larger role in SIGGRAPH-related research than it seems to now. Why? What was and is the role of computer graphics research in the larger scope of user experience, and vice versa?
- What were the main questions and challenges in the evolution of user experience in computer graphics you faced throughout your career? Still face?
- Tales from the trenches: our panelists tell us some of the unexpected things that happened in their work in this evolution? What surprised and/or delighted them;
- Finally, our panelists discuss the torch to pass on. If they had to build a handbook for young researchers in CG and interaction, what would the most important aspects be? What are the key parts of their experience they hope would help

young researchers and practitioners in their career paths to further the field?

### 3 OUR PANELISTS

**James Foley** is a computing researcher, innovative educator and respected author in both computer graphics and human-computer interaction, one of the SIGGRAPH pioneers who led the establishment of human-computer interaction as a major discipline. He is Professor and Fleming Chair Emeritus in the Georgia Tech College of Computing. He co-authored computer graphics texts, is a Member of NAE and Fellow of AAAS, ACM and IEEE. He received ACM/SIGGRAPH's Lifetime and ACM/SIGCHI's Lifetime Achievement Awards. The GVU Center, which he founded, ranked number one in 1996 US News and World Report for graduate computer science work in graphics & user interaction. From 1996-99 he directed Mitsubishi Electric's US Research Labs; from 2001 to 2005 he chaired the Computing Research Association - an organization of 250 computer science departments, professional societies and research labs. In addition to several hundred peer-reviewed articles and books, he is the co-author of several widely used textbooks in the field of computer graphics, notably *Computer Graphics: Principles and Practice*, with over 400,000 copies sold in ten languages.

**Gudrun Klinker** studied computer science (informatics) in Erlangen, Hamburg and at Carnegie-Mellon University in Pittsburgh, focusing on computer vision. In 1989, she joined the visualization group of DEC-CRL to develop a reusable tele-collaborative data exploration environment for the visualization of high-dimensional data. Since 1995, she has been researching various aspects of AR and VR, at ECRC, Fraunhofer FHG-IGD, and since 2000 at the TU Munich. Here, her research focus lies on developing approaches to ubiquitous augmented reality that lend themselves to realistic industrial applications, as well as to gamified educational purposes. Professor Klinker is a member of the steering committee of the IEEE Virtual Reality conference. She has served on numerous program committees such as IEEE ISMAR, ACM VRST, IEEE 3DUI, and ACM UIST. She is author and co-author of more than 220 peer-reviewed scientific publications, cited by more than 9000 peer-reviewed articles.

**Kiyoshi Kiyokawa** is Professor at the Nara Institute of Science and Technology (NAIST) where he leads the Cybernetics and Reality Engineering (CARE) Laboratory. His pioneering work began in immersive modeling, object manipulation and interactive techniques. Kiyokawa's interests began to gradually shift from AR/VR to human augmentation and assistive interfaces after he compiled the VR Technology Roadmap for the Virtual Reality Society of Japan in 2013, which envisions a future in which XR technology can increase people's options for living and realize an inclusive society. Since then, he has frequently worked on vision augmentation and correction projects. He has also developed various multimodal interfaces, including one of the first systems to use drone-based midair haptics in 2013 and a GAN-based food-to-food translation system for taste modulation in 2018.

**Maureen Stone** is most recently the Director of Research at Tableau Software, and has a long career of contributions in both theoretical and applied contexts. She is perhaps best known for her

expertise in digital color and its applications, but has broad experience in digital design, interactive graphics and user interface design. She began her career working on projects related to the then new fields of desktop publishing, digital document design and digital printing, pivoting to the exploding domain of data visualization and the design of techniques, tools and representations focused on the interaction between human cognition and visual design. She is the author of the seminal book *A Field Guide to Digital Colour* and of numerous papers addressing how humans process and interact with visual representations from both theoretical and applied perspectives.

**Lyn Bartram (Moderator)** is Professor in the School of Interactive Art and Technology at SFU and Director of the Vancouver Institute of Visual Analytics, an SFU Research Institute engaging researchers, practitioners and organizations with challenges and opportunities in the emerging universe of big data. Her work explores the intersecting potential of interactive technologies, visual analytics, computer graphics and computational media from both theoretical and applied perspectives, particularly to better support data-enabled thinking beyond the traditional applications of expert data science.